

MAJOR MOON

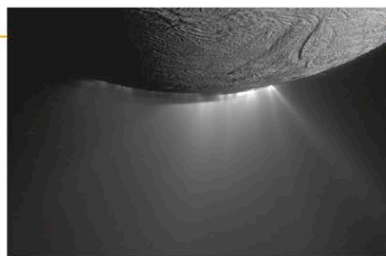
Enceladus

DISTANCE FROM SATURN 147,898 miles
(238,020 km)

ORBITAL PERIOD 1.37 Earth days

DIAMETER 313 miles (504 km)

Enceladus orbits within the broad E ring of Saturn. Its orbit lies within the densest part of the ring, which suggests that Enceladus could be supplying the ring with material. The moon is in synchronous rotation with Saturn. The frosty surface of Enceladus is highly reflective and makes this moon particularly bright, the brightest in the solar system. The surface terrain suggests that this frigid moon has experienced a long history of tectonic activity and resurfacing. The extent of the geological change is surprising in



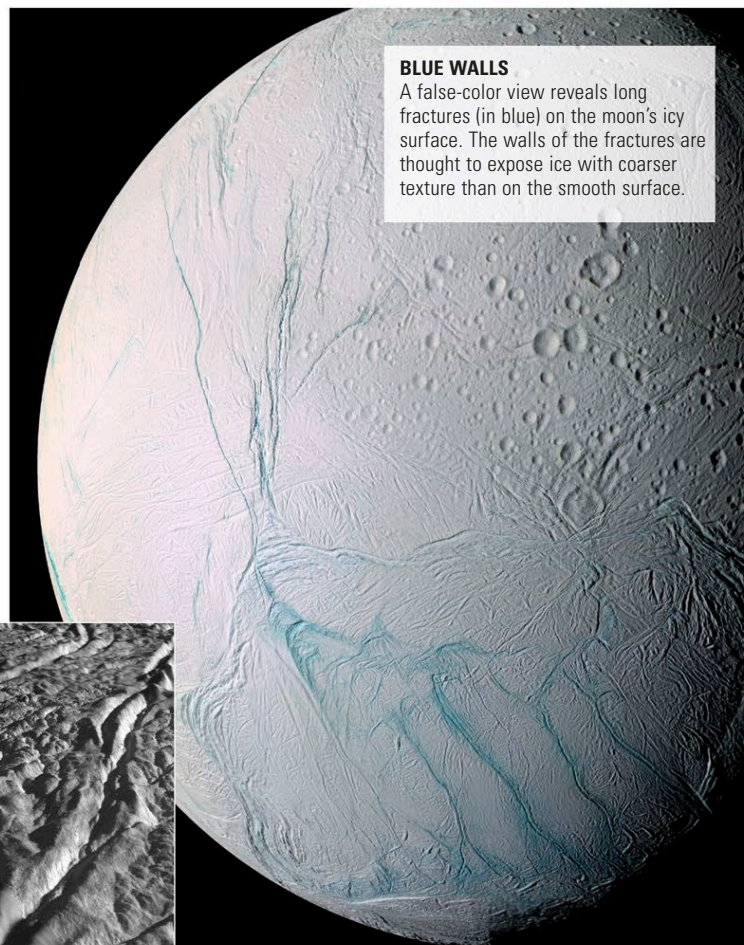
WATER JETS

Ice and water vapor spray out from so-called tiger stripes near the south pole of Enceladus, as shown here in this image from the Cassini probe.

such a small world—Mimas (see p.193) is about the same size but is inactive. Craters are concentrated in some regions, and elsewhere there are grooves, fractures, and ridges. Images processed to accentuate color differences have revealed previously unseen detail. The blue color seen in some fracture walls could be due to the exposure of solid ice or because the composition or size of particles in the buried ice is different from that on the surface. Enceladus was discovered by William Herschel on August 28, 1789.

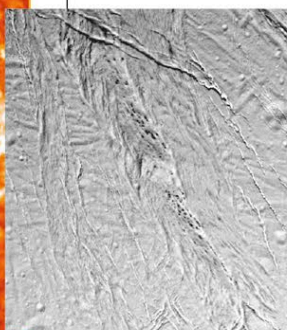
BAGHDAD SULCUS

This is a close-up of part of Baghdad Sulcus, the longest of several linear structures popularly termed tiger stripes in the south polar region of Enceladus.



BLUE WALLS

A false-color view reveals long fractures (in blue) on the moon's icy surface. The walls of the fractures are thought to expose ice with coarser texture than on the smooth surface.



SMOOTH PLAINS

This region of smooth plains has a band of chevron-shaped features running across its center, cut across at the top by a system of crevasses.

INNER MOON

Telesto

DISTANCE FROM SATURN 183,093 miles
(294,660 km)

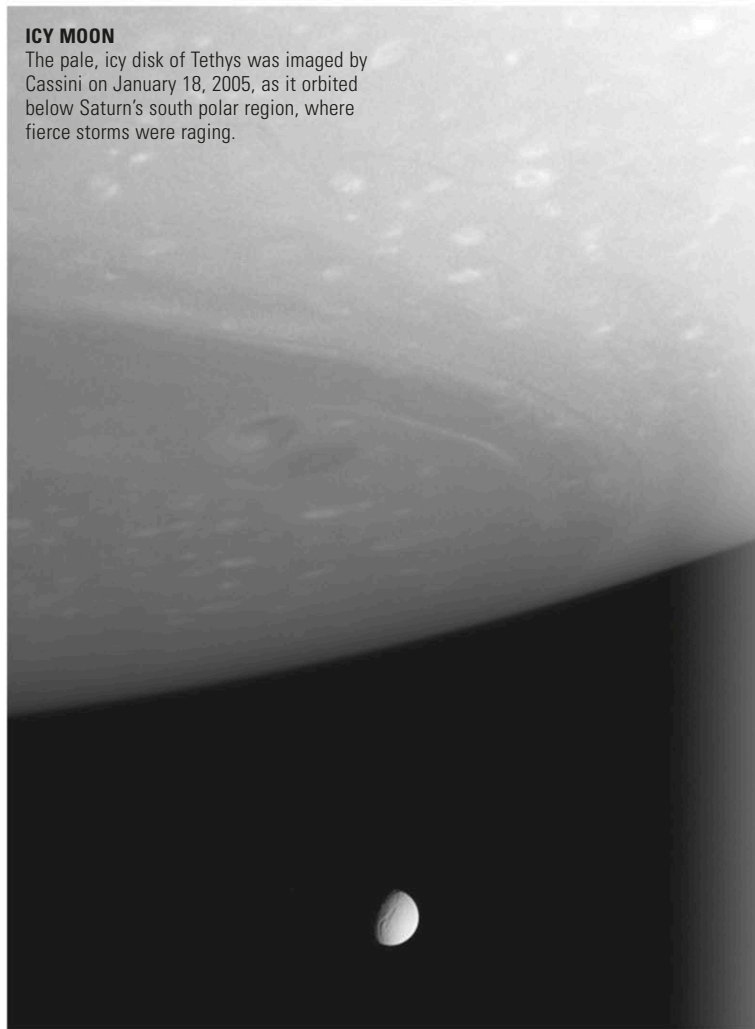
ORBITAL PERIOD 1.89 Earth days

LENGTH 20 miles (32.5 km)

Telesto shares an orbit within the E ring with two other moons: Calypso, which is about the same size as Telesto, and the much larger Tethys (right). Telesto moves along the orbit 60° ahead of Tethys, and Calypso follows 60° behind Tethys. The positions taken on the orbit by the two smaller moons are called the Lagrange points. In these positions, the two small moons can maintain a stable orbit balanced between the gravitational pull of Saturn and that of Tethys. Telesto and Calypso were discovered in 1980—Calypso by Earth-based observation and Telesto in Voyager images. The probe revealed two irregularly shaped moons.

ICY MOON

The pale, icy disk of Tethys was imaged by Cassini on January 18, 2005, as it orbited below Saturn's south polar region, where fierce storms were raging.



MAJOR MOON

Tethys

DISTANCE FROM SATURN 183,093 miles
(294,660 km)

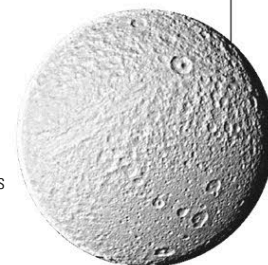
ORBITAL PERIOD 1.89 Earth days

DIAMETER 660 miles (1,062 km)

The Italian-French astronomer Giovanni Cassini discovered Tethys on March 21, 1684. Nearly 300 years later, it was discovered that Tethys shares its orbit with two far smaller moons: Telesto (left) and Calypso. Its surface shows that Tethys has undergone tectonic change and resurfacing. Two features stand out. A 248-mile (400-km) wide impact crater called Odysseus dominates the leading hemisphere. Large but shallow, its original bowl shape has been flattened by ice flows. The second large feature is the Ithaca Chasma on the side of Tethys facing Saturn. This vast canyon system extends across half of the moon. It may have been formed by tensional fracturing as a result of the impact that produced the Odysseus Crater, or when Tethys's interior froze and the moon expanded in size and stretched its surface.

ITHACA CHASMA

This canyon system, which is up to 2.5 miles (4 km) deep, runs from the lower left of the prominent Telemachus Crater (top right).



SMOOTH MOON

Telesto's surface appears less cratered than most of Saturn's other moons in this image taken by Cassini from a distance of only 9,000 miles (14,500 km).